Differential Equations and Operators

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Overview

- Incorporating derivatives permits a model for the simultaneous variation in function and one or more of its derivatives.
- Can we use mixtures of derivatives instead of simple derivatives?
- Can we use derivatives on the covariate side, the response side, or both?
- Can we extend our smoothing to these settings?
- We need a review of DEQs first.

A simple linear differential equation

- Goal: to model the behavior of the output function x of a system and its derivatives without considering any external influences.
- A *homogeneous* linear differential equation:

$$Dx(t) = -\beta x(t).$$

- Dependent variable: the derivative of output x.
- Independent variables: x itself.
- $-\beta$: "$-$" since most real-life systems have a natural tendency to return to their resting state.
- Solution:

$$x(t) = Ce^{-\beta t}.$$

A general form and its solution

- A general *homogeneous* linear differential equation with constant coefficients has the form:

$$\sum_{j=0}^{m-1} \beta_j D^j x(t) + D^m x(t) = 0.$$

- To obtain the solution to this equation, first solve its *characteristic equation*:

$$z^m + \sum_{j=0}^{m-1} \beta_j z^j = 0$$

- Denote these complex roots by $z_j, j = 1, \ldots, m$.
- The solution to the differential equation is: $\sum_{j=1}^{m} C_j e^{z_j t}$, where C_j's are constants.

An input-output system

- Setting: An input/output system has an input function u that in some way modifies an output function x.
- A *nonhomogeneous* linear differential equation:

$$Dx(t) = -\beta x(t) + \alpha u(t).$$

- Dependent variable: the derivative of output x.
- Independent variables: x itself and an input function u.
- αu is called a *forcing function*.
- Solution:

$$x(t) = Ce^{-\beta t} + \alpha \int_0^t e^{-\beta(t-s)} u(s) ds.$$

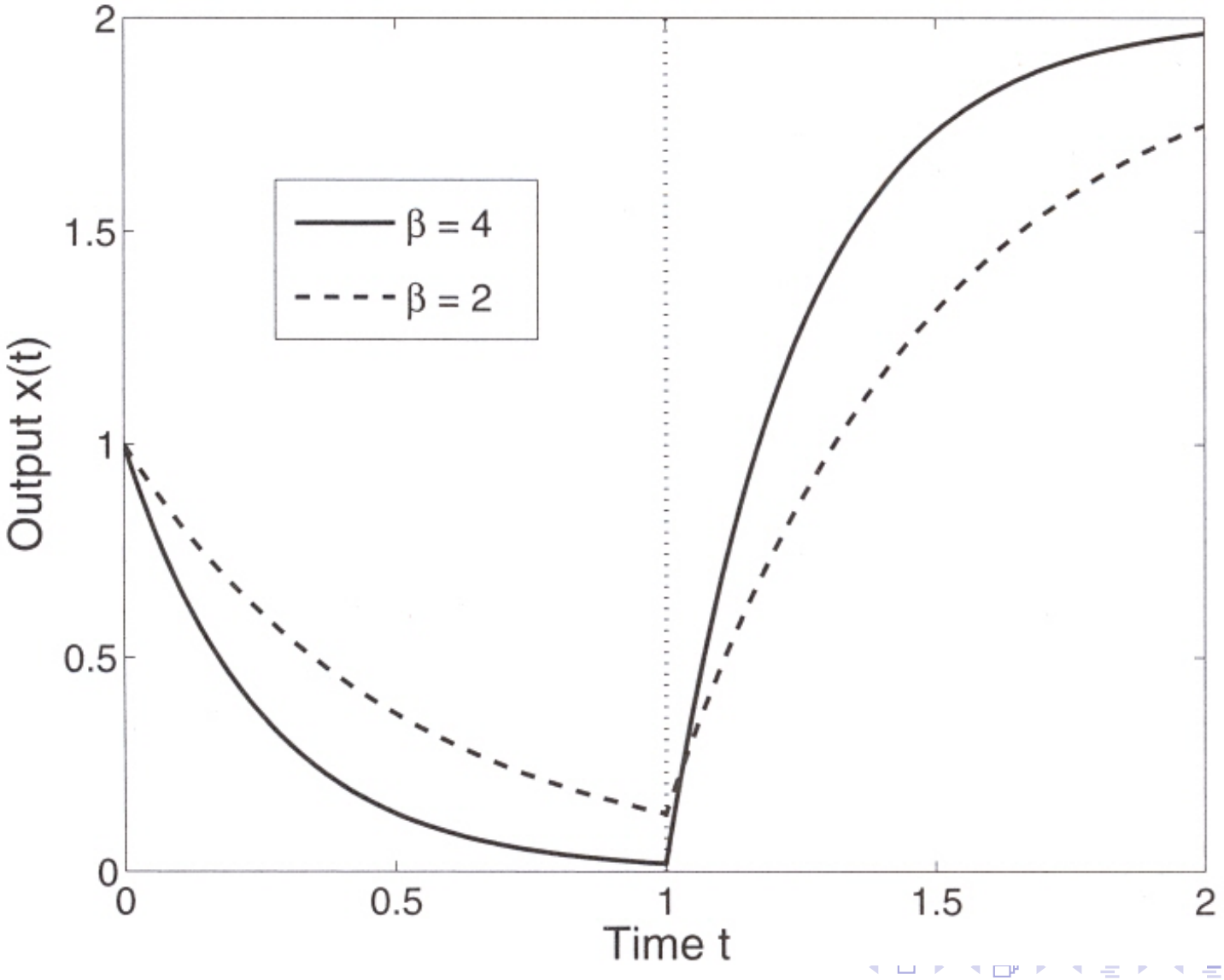
A system with a step input

- Consider an input function: $u(t) = 0$ for $0 \leq t \leq 1$ and $= 1$ for $t > 1$.
- Also, let's set $C = x(0) = 1$.
- The solution:

$$\begin{aligned} x(t) &= e^{-\beta t}, 0 \leq t \leq 1, \\ &= \alpha/\beta + [1 - (\alpha/\beta)e^{\beta}]e^{-\beta t}, t > 1. \end{aligned}$$

- β: the rate of change in x in response to a step change in u.
- α/β: the *gain* of the system.
- α thus controls the gain of the system given a β.

Plot of solution with $\alpha/\beta = 2$



The corresponding penalty

- Define $L = \beta I + D$.
- The equation becomes

$$Lx(t) = \alpha u(t) + \epsilon(t).$$

- The roughness penalty

$$\text{PEN}(x) = \int [Lx(t)]^2 dt$$

penalizes the failure of x to satisfy the differential equation $Dx = -\beta x$ corresponding to the operator L.

Why nonconstant coefficients?

- Many systems can increase more rapidly than they decrease, or vice versa.
- A constant rate of change β is not sufficient.
- Need a time-varying rate of change $\beta(t)$
- Similarly, need a time-varying $\alpha(t)$ in the force.

A first-order DEQ with nonconstant coefficients

- A first-order DEQ with nonconstant coefficients:

$$Dx(t) = -\beta(t)x(t) + \alpha(t)u(t).$$

- The solution is

$$x(t) = Cx_0(t) + \int_0^t \alpha(s)u(s)x_0(t)/x_0(s)ds,$$

where $x_0(t) = \exp[-\int_0^t \beta(s)ds]$.

- The functional ratio $x_0(t)/x_0(s)$ in the integral defines the *Green's function* for the differential equation.

Higher order equations

- A higher order DEQ with nonconstant coefficients:

$$D^m x(t) = -\sum_{j=0}^{m-1} \beta_j(t) D^j x(t) + \alpha(t) u(t).$$

- This is needed when there are more than two times scales for the events in the system.
- The solution is much more complicated.

A higher-order DEQ example

- A second-order DEQ:

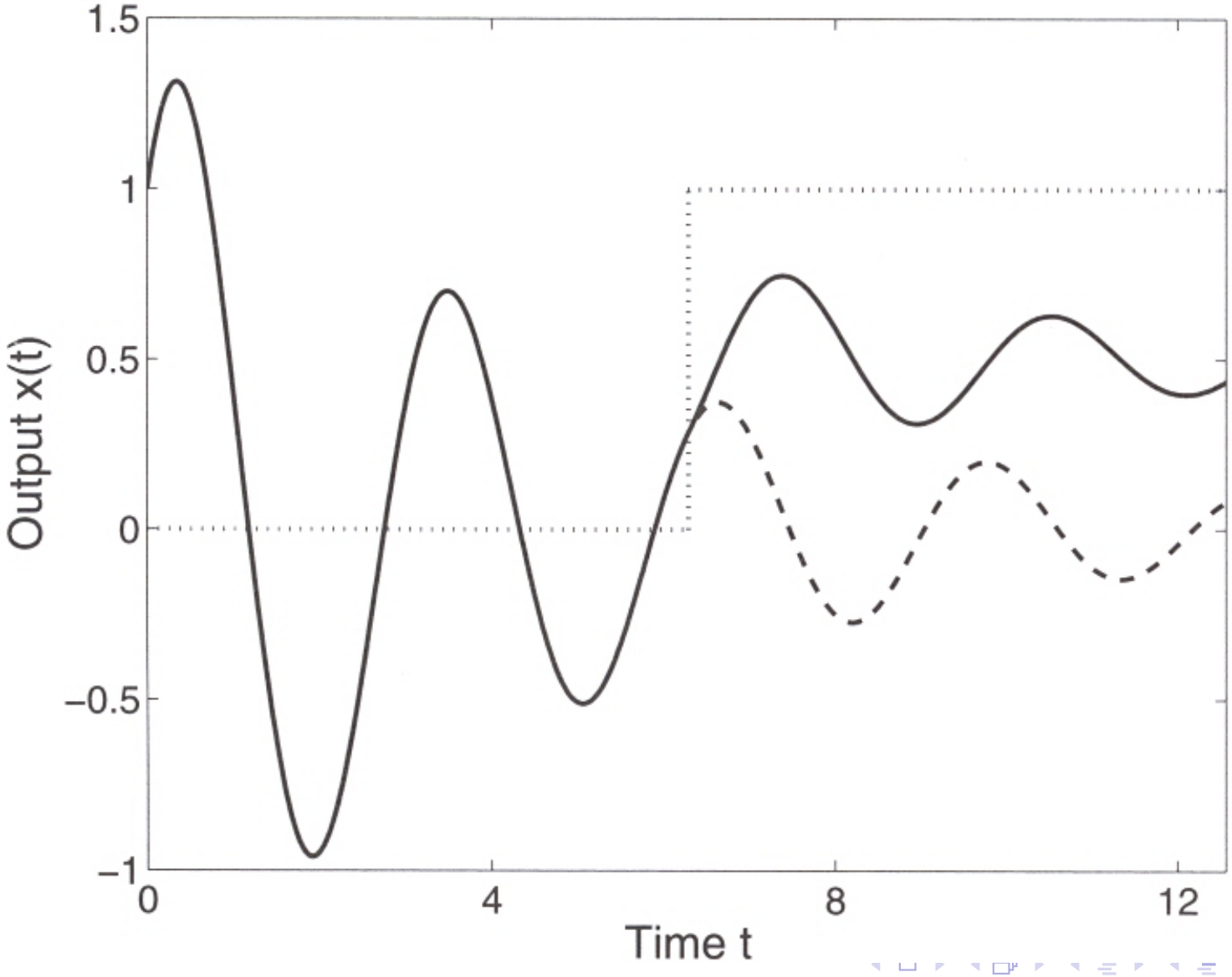
$$D^2x(t) = -4.04x(t) - 0.4Dx(t) + 2u(t),$$

  where the forcing function $u(t)$ steps from 0 to 1 at time $t = 2\pi$.
- A solution to the homogeneous equation is

$$x_0(t) = e^{-0.2t}[\sin(2t) + \cos(2t)].$$

- The solution to the nonhomogeneous equation is complicated.
- See a plot of the solutions.

Plot of solutions to the 2nd-order DEQ



Three time scales in the solution

- The longest scale trend: The overall oscillation level is first about 0 and then later (after $t = 2\pi$) about 0.5.
- The medium scale trend: the exponential decay in the amplitude of the oscillation.
- The shortest scale trend: the oscillation with period π.

A handwriting example

Ramsay (2000) observes features in script at four time scales:

- The overall spatial position of the script (the line on which it is written): requiring some considerable seconds per line.
- The movement of the script from left to right within a line: taking place over several seconds.
- The strokes and loops within the script: produced about eight times a second.
- The sharper transient effects due to the pen striking or leaving the paper: lasting of the order of 10 ms.

Systems of DEQs

- A feedback system: a loop for the process permitting the output x to influence the input u.
- Example: a doctor adjusts the medication u of the patient according to changes in the symptoms x.
- A DEQ system with two components

$$Dx(t) = -\beta_x(t)x(t) + \alpha_x(t)u(t),$$
$$Du(t) = -\beta_u(t)x(t) + \alpha_u(t)u(t).$$

Higher order DEQs to DEQ systems

- A differential equation of order m can be expressed as a system of m first-order equations.
- Consider a second-order equation:

$$D^2x(t) = -\beta_0(t)x(t) - \beta_1(t)Dx(t).$$

- After defining $y(t) = Dx(t)$, an equivalent DEQ system with two components is

$$\begin{aligned} Dx(t) &= y(t), \\ Dy(t) &= -\beta_1(t)y(t) + \beta_0(t)x(t). \end{aligned}$$

Producing new functional observations

- Newton's law $a(t) = F(t)/M$: the application of some force $F(t)$ at time t on an object with mass M has an **immediate** impact on acceleration $a(t)$.
- Force has an indirect impact on velocity: $v(t) = v_0 + M^{-1} \int_0^t F(u)du$.
- and an even less direct impact on what we directly observe, namely position: $s(t) = s_0 + v_0 t + M^{-1} \int_0^t \int_0^u F(z)dzdu$.
- The world we experience is two integrals away from reality!
- Other "non-dynamic" examples of F: release of adrenal hormones during puberty, a muscle contraction with respect to certain part of the body.

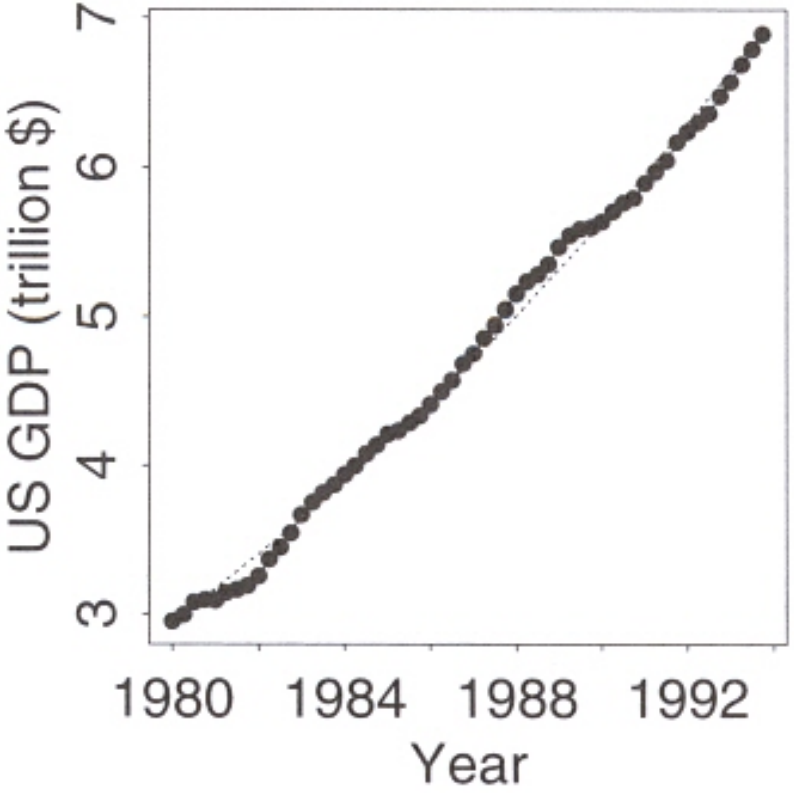
GDP data: some background

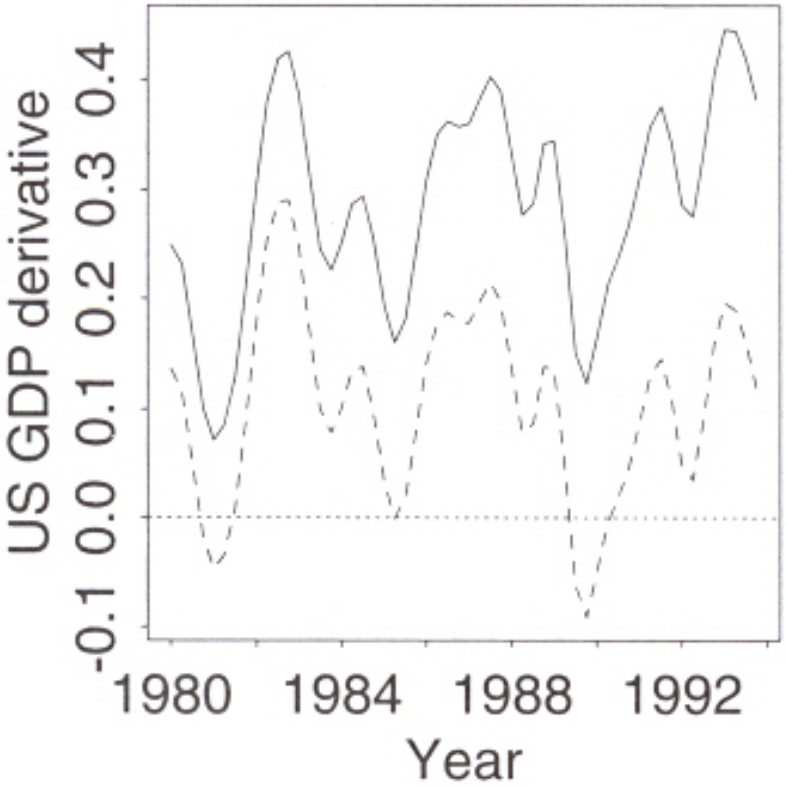
- The gross domestic product (GDP) of a county is the financial value of all goods and services produced in that country.
- GDP tends to exhibit a percentage change each year in stable times.
- There can be considerable fluctuation in such changes from year to year.
- Over long periods the fluctuations tend to even out and the long-range trend in GDP tends to be roughly exponential.

GDP data description

- Data: quarterly GDP values for 15 countries in the Organization for Economic Cooperation and Development (OECD) for the years 1980 through 1994.
- The values of GDP are expressed in its own currency, and thus scales are not comparable across countries.
- The reports from some countries contain strong seasonal effects.
- Others smooth them out before reporting.

The GDPs of USA





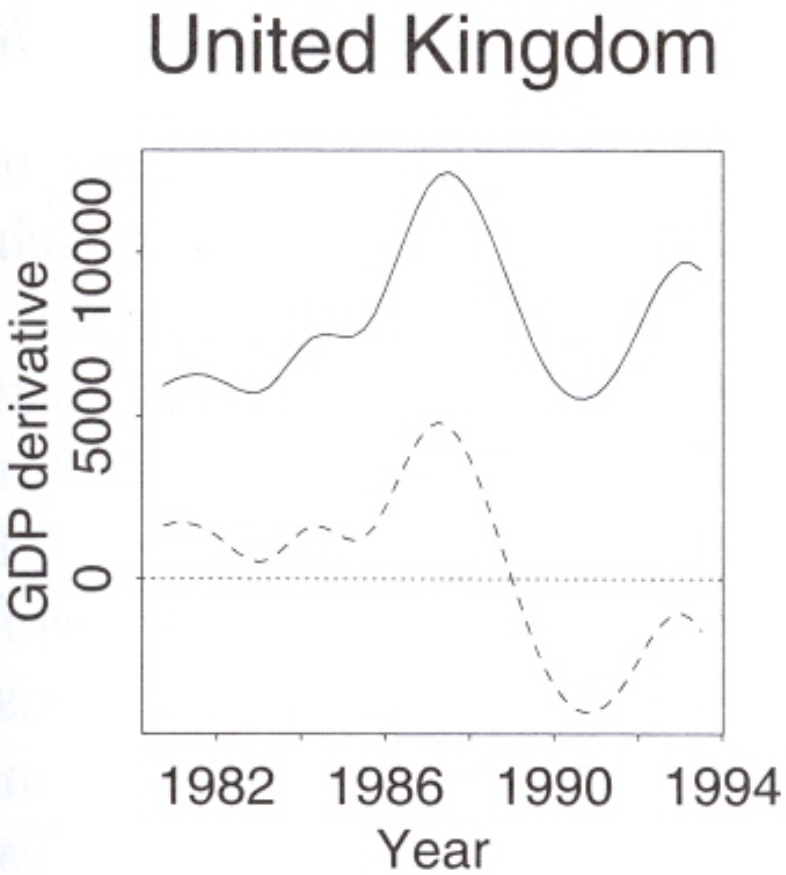
The GDPs of USA

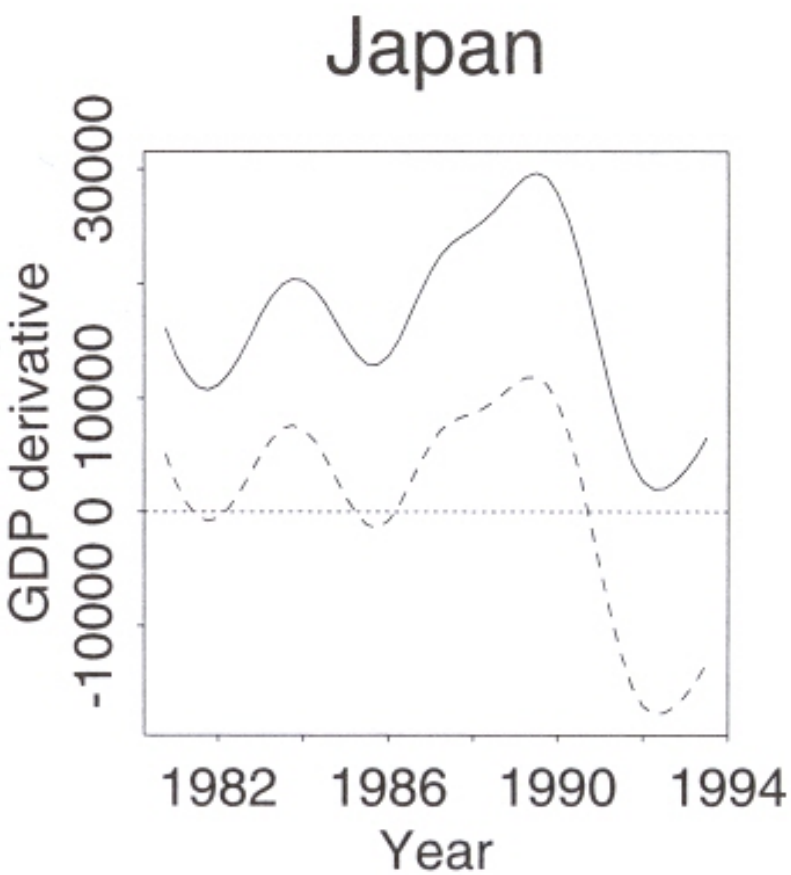
- Hardly visible seasonal trend.
- A smooth fit is computed using a penalty on D^4GDP.
- DGDP (solid line in right panel) shows rapidly advancing economy in 1983, 1987, and 1993, as well as slowdowns in 1981, 1985, and 1990.

Exponential growth for the GDPs of USA

- The best-fitting exponential trend is $C\exp(\gamma t)$ with $\gamma = 0.038$: the US economy tended to grow at about 4% per year over the period of 1980-1994.
- Define $Lx = -\gamma x + Dx$ such that the solution to $Lx = 0$ is an exponential growth function.
- LGDP (dashed line in right panel) shows departures from 0: there are three cycles of shorter term growth in GDP that depart from the long-term exponential trend.

The GDPs of UK and Japan





The GDPs of UK and Japan

- UK had only one boom period, followed by a recession and an uncertain recovery after the recession.
- Japan had two boom periods, with the later and bigger boom period followed by a deep and late recession.

Differential operators to regularize models

- Consider nonparametric regression model

$$y_i = x(t_i) + \epsilon_i, \quad i = 1, \ldots, n,$$

where x is the unknown smooth regression function of interest.

- Consider estimating x via minimizing the penalized least squares criterion

$$\text{PENSSE}_\lambda(x) = n^{-1} \sum_{i=1}^{n} [y_i - x(t_i)]^2 + \lambda \int (Lx)^2(t) dt$$

for some differential operator L.

Estimating L?

- With a properly chosen λ, such an estimate will have small bias as well as much lower variance and thus smaller integrated mean squared error (IMSE) since
$$\text{IMSE}(\hat{x}) = \text{Bias}^2(\hat{x}) + \text{Var}(\hat{x}).$$
- L annihilates functions that satisfy $Lx = 0$.
- Can we use the information in N replications x_i of functional observations to **estimate** an operator L that comes close to satisfying $Lx_i = 0$?
- Then we can use this information to improve the smoothing.

Differential operators to partition variation

- The *null space* of a differential operator L is

$$\ker(L) = \{x : Lx = 0\},$$

  often a finite-dimensional space.
- Hence L splits x into two parts: $x - Lx$ that can be expressed in terms of a linear combination of the null space basis functions ξ_j and Lx that is the orthogonal complement.
- This decomposition also splits up the corresponding functional variation.

The orthonormal partition of variation

- Another way of partition uses a set of orthonormal basis functions ϕ_k, e.g., the Fourier basis functions.
- Project x to the space spanned by the first m basis functions:

$$Px = \sum_{k=1}^{m} c_k \phi_k.$$

- The complementary project operator $Q = I - P$ has analogous properties to L.

Comparing L and Q: An Example

Consider the Fourier expansion

$$x(t) = c_0 + \sum_{k=1}^{\infty} [c_{2k-1} \sin(2k\pi t) + c_{2k} \cos(2k\pi t)].$$

Design L and Q to eliminate the first three terms of the expansion, namely, $c_0 + c_1 \sin(2\pi t) + c_2 \cos(2\pi t)$:
For the projection operator Q, it is straightforward to see

$$Qx(t) = \sum_{k=2}^{\infty} [c_{2k-1} \sin(2k\pi t) + c_{2k} \cos(2k\pi t)].$$

Comparing L and Q Example: Deriving L

To construct L,

- eliminating c_0 need the first-order derivative D,
- eliminating $c_1 \sin(2\pi t) + c_2 \cos(2\pi t)$ requires $D^2 + 4\pi^2 I$,
- so $Lx(t) = D^3 x(t) + 4\pi^2 D x(t)$, and

$$Lx(t) = \sum_{k=2}^{\infty} 8\pi^3 k(k^2 - 1)[-c_{2k-1} \cos(2k\pi t) + c_{2k} \sin(2k\pi t)].$$

Comparing L and Q

$$Qx(t) = \sum_{k=2}^{\infty} [c_{2k-1} \sin(2k\pi t) + c_{2k} \cos(2k\pi t)].$$

$$Lx(t) = \sum_{k=2}^{\infty} 8\pi^3 k(k^2 - 1)[-c_{2k-1} \cos(2k\pi t) + c_{2k} \sin(2k\pi t)].$$

- If Q or L is used in penalty, L puts more penalty on high-frequency terms in the Fourier expansion.
- High-frequency terms are often noises!

Finding a linear differential operator

- Suppose we have a set of functions $\{\xi_1, \ldots, \xi_m\}$ in mind, and want to find a linear differential operator L that annihilates these functions.
- An example: find an order-2 operator $L = \beta_0(t)I + \beta_1(t)D + D^2$ that can annihilate signals of the form

$$x(t) = A(t)[c_1 \sin(\omega t) + c_2 \cos(\omega t)].$$

- First, we need an invertible *Wronskian matrix*

$$\mathbf{W}(t) = [\boldsymbol{\xi}(t) \quad D\boldsymbol{\xi}(t) \quad \cdots \quad D^{m-1}\boldsymbol{\xi}(t)].$$

- Second, we will look for an order-m linear operator with weight functions $\boldsymbol{\beta} = (\beta_0(t), \ldots, \beta_{m-1}(t))'$ such that the $\boldsymbol{\beta}$ solves the system of m linear equations

$$\mathbf{W}(t)\boldsymbol{\beta}(t) = -D^m\boldsymbol{\xi}(t).$$

An Example

- Find $L = \beta_0(t)I + \beta_1(t)D + D^2$ to annihilate signals

$$x(t) = A(t)[c_1 \sin(\omega t) + c_2 \cos(\omega t)].$$

- Let $\mathrm{S}(t)$ stand for $\sin(\omega t)$ and $\mathrm{C}(t)$ for $\cos(\omega t)$, then $\xi_1 = A\mathrm{S}$, $\xi_2 = A\mathrm{C}$ and the Wronskian matrix is

$$\mathbf{W} = [\boldsymbol{\xi} \quad D\boldsymbol{\xi}] = \begin{bmatrix} A\mathrm{S} & (DA)\mathrm{S} + \omega A\mathrm{C} \\ A\mathrm{C} & (DA)\mathrm{C} - \omega A\mathrm{S} \end{bmatrix}$$

with $|\mathbf{W}| = -\omega A^2$.

- The weight functions are

$$\boldsymbol{\beta} = -\mathbf{W}^{-1} D^2 \boldsymbol{\xi} = \begin{bmatrix} \omega^2 + 2(DA/A)^2 - D^2 A/A \\ -2DA/A \end{bmatrix}.$$

- The annihilating operator is

$$Lx = [\omega^2 + 2(DA/A)^2 - D^2 A/A]x - 2(DA/A)(Dx) + D^2 x.$$

Finding the functions in ker(L)

- A reverse problem: given a linear differential operator L of order m, we want to identify m linearly independent solutions ξ_j to the homogeneous equation $Lx = 0$.
- A variety of methods dealing with this.
- Refer to any introductory DEQ book.

Boundary conditions and constraints

- Consider an order-m linear differential operator L.
- The solution to the equation $Lx = 0$ is a function space of dimension m.
- To uniquely determine a solution function, we need m additional conditions, often called *initial conditions* or *boundary conditions*.
- An example: Find a function x such that

$$\beta Dx + D^2x = 0 \text{ and } x(0) = 0, Dx(0) = 1.$$

Constraint operator B

- **Initial Operator**

$$B_0 x = [x(0) \quad Dx(0) \quad \cdots \quad D^{m-1}x(0)]'.$$

- **Boundary Operator**

$$B_B x = [x(0) \quad x(T) \ \cdots \ D^{(m-2)/2}x(0) \quad D^{(m-2)/2}x(T)]'.$$

- **Periodic Operator**

$$B_P x = [x(T) - x(0) \quad Dx(T) - Dx(0) \quad \cdots D^{(m-1)}x(T) - D^{(m-1)}x(0)]'.$$

- **Integral Operator**

$$B_I x = [\int \xi_1(t)x(t)dt \quad \cdots \quad \int \xi_m(t)x(t)dt]',$$

where $\xi_1, \ldots, \xi_m$ are m linearly independent weight functions.

Inner product defined by L and B

- Consider the case when $\ker B \cap \ker L = 0$, that is, the two function spaces $\ker B$ and $\ker L$ complements each other.
- We can define an inner product of two functions

$$\langle x, y \rangle_{B,L} = (Bx)'(By) + \int (Lx)(t)(Ly)(t)dt.$$

- This introduces a norm

$$\|x\|_{B,L}^2 = (Bx)'(Bx) + \int (Lx)^2(t)dt.$$

Polynomial smoothing splines

- Recall that a polynomial smoothing spline can be defined as the minimizer of

$$\frac{1}{n}\sum_{i=1}^{n}(y_i - x(t_i))^2 + \lambda \int_0^1 (D^m x)^2(t)dt$$

in the function space $\mathcal{C}^{(m)}[0,1] = \{f : f^{(m)} \in \mathcal{L}^2[0,1]\}$.
- Here $L = D^m$, and we can have two versions of B.
- First version: an initial operator B introduces

$$\langle f, g\rangle_{B,L} = \sum_{\nu=0}^{m-1} D^\nu f(0) D^\nu g(0) + \int (D^m f)(D^m g).$$

- Second version: an integral operator B introduces

$$\langle f, g\rangle_{B,L} = \sum_{\nu=0}^{m-1} (\int_0^1 D^\nu f)(\int_0^1 D^\nu g) + \int (D^m f)(D^m g).$$

DEQs and Operators

- Have reviewed on general solutions to linear differential equations.
- Using differential operator to create new functional objects of interest.
- Differential operator partitions the variation and puts heavier penalty on high frequency terms.
- Find a linear differential operator to annihilate functions of special forms.
- Conditions/constraints on DEQs.